

Ethan Shen

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Education

University of Washington

B.S. in Computer Science, B.A. in Mathematics, Minor in History

Seattle, WA

Aug. 2022 - Present

- **GPA:** 3.96 / 4.00
- **Coursework:** Reinforcement Learning (Grad), Deep Learning (Grad), Natural Language Processing (Grad), Operating Systems, Software Design, Data Structures/Parallelism, Linear Algebra I and II, Probability I and II, Multivariate Calculus
- **Activities:** Research Assistant, Teaching Assistant, CSE Faculty Hiring Interviewer, CSE Student Advisory Council, Swim Club

Skills

Languages Java, Python, SQL, JavaScript, C, C++, HTML/CSS
Frameworks PyTorch, Jax, HuggingFace, React, Node.js, NextJS, Flask, JUnit, scikit-learn
Dev Tools Git, Google Cloud Platform, AWS, Azure, MongoDB, DynamoDB, Docker, Linux

Publications

Perception Tokens Enhance Visual Reasoning in Multimodal Language Models

Mahtab Bigverdi, Zelun Luo, Cheng-Yu Hsieh, **Ethan Shen**, Dongping Chen, Linda Shapiro, Ranjay Krishna.
CVPR 2025.

Superposed Decoding: Multiple Generations from a Single Autoregressive Inference Pass.

Ethan Shen, Alan Fan, Sarah M Pratt, Jae Sung Park, Matt Wallingford, Sham M Kakade, Ari Holtzman, Ranjay Krishna, Ali Farhadi, Aditya Kusupati.
NeurIPS 2024.

Are "Hierarchical" Visual Representations Hierarchical?

Ethan Shen, Ali Farhadi, Aditya Kusupati.
Workshop on Symmetry and Geometry in Neural Representations @ NeurIPS 2023.

Experience

Allen Institute of Artificial Intelligence

Research Intern (Prof. Tim Dettmers)

Seattle, WA

Feb. 2025 - Present

- Leading development of an open-source alternative to closed-source coding agents such as Claude Code.
- Experimented with network stacking for low-cost LLM scaling and implemented distributed training for 100M to 7B parameter models on a trillion-token scale.

RAIVN Lab @ UW

AI Research Assistant (Prof. Ali Farhadi, Prof. Ranjay Krishna)

Seattle, WA

Jun. 2023 - Present

- Pioneered the use of latent token representations for multimodal reasoning in vision-language models like Qwen and LLaVA.
- Created LLM decoding algorithm (in use at Meta) improving inference efficiency and accuracy for multi-draft generation.
- Developed suite of hierarchical vision datasets to measure the performance of image classification models on understanding complex visual hierarchies.

Amazon

Software Engineer Intern

Seattle, WA

Jun. 2024 - Sep. 2024

- Built a scalable GenAI pipeline with AWS Bedrock, Lambda, and DynamoDB to automatically process up to 15,000 financial documents a month, saving over 1200 hours of manual work per year.
- Created an internal invoice dataset of 45+ vendors and improved pipeline accuracy to 96% through prompt engineering, fine tuning, and heuristic filtering.

Papyrus

Machine Learning Engineer Intern

San Francisco, CA

Dec. 2023 - Mar. 2024

- Worked at pre-seed startup and developed first-of-its-kind feature to label speakers in transcripts using long-context knowledge from language models.
- Built evaluation pipeline for transcription/translation using NextJS and Python and deployed it with AWS Lambda and Batch, saving 20 hours of testing monthly.

Sensor Systems Lab @ UW

Robotics Research Assistant (Prof. Joshua Smith)

Seattle, WA

Jun. 2022 - Jun. 2023

- Designed and built an acoustic levitator, a tool that uses ultrasonic sound to levitate fragile objects for scientific experiments.
- Programmed mathematical algorithms in Python and C++ to simulate, predict, and control the rotation of objects in the levitator, with a 15x speedup compared to existing simulations.

Professional Services

2025	NeurIPS 2025 Reviewer
2024	EMNLP 2024 Reviewer
2024	UW CSE: Student Interviewer for Faculty Hiring
2023	UW CSE: Teaching Assistant
2022-2023	UW CSE: Student Advisory Council Officer

Presentations

Oct. 2024	UW RAIVN Lab (Host: Ali Farhadi) , Topic: Superposed Decoding
Aug. 2024	AWS AI Labs CodeGen (Host: Zijian Wang) , Topic: Superposed Decoding
Aug. 2024	Amazon LLM Reasoning Group (Host: Linbo Liu) , Topic: Superposed Decoding
Jul. 2024	AWS AI Labs Quicksight (Host: Patrick Ng) , Topic: Superposed Decoding

Honors & Awards

2023	NeurIPS Travel Award , Conference for Neural Information Processing Systems
2022	FEEA Merit Scholarship , US Department of Veterans Affairs
2021	42/45 Score (Top 8% Internationally) , IB Diploma Programme
2021	National Merit Finalist (Top 1% Nationally) , National Merit Program